

AVL Tree Properties | Problems on AVL Tree

Data Structures

AVL Tree-

Before you go through this article, make sure that you have gone through the previous article on [AVL Trees](#).

We have discussed-

- AVL trees are self-balancing binary search trees.
- In AVL trees, balancing factor of each node is either 0 or 1 or -1.

In this article, we will discuss AVL Tree Properties.

AVL Tree Properties-

Important properties of AVL tree are-

Property-01:

Maximum possible number of nodes in AVL tree of height H

$$= 2^{H+1} - 1$$

Example-

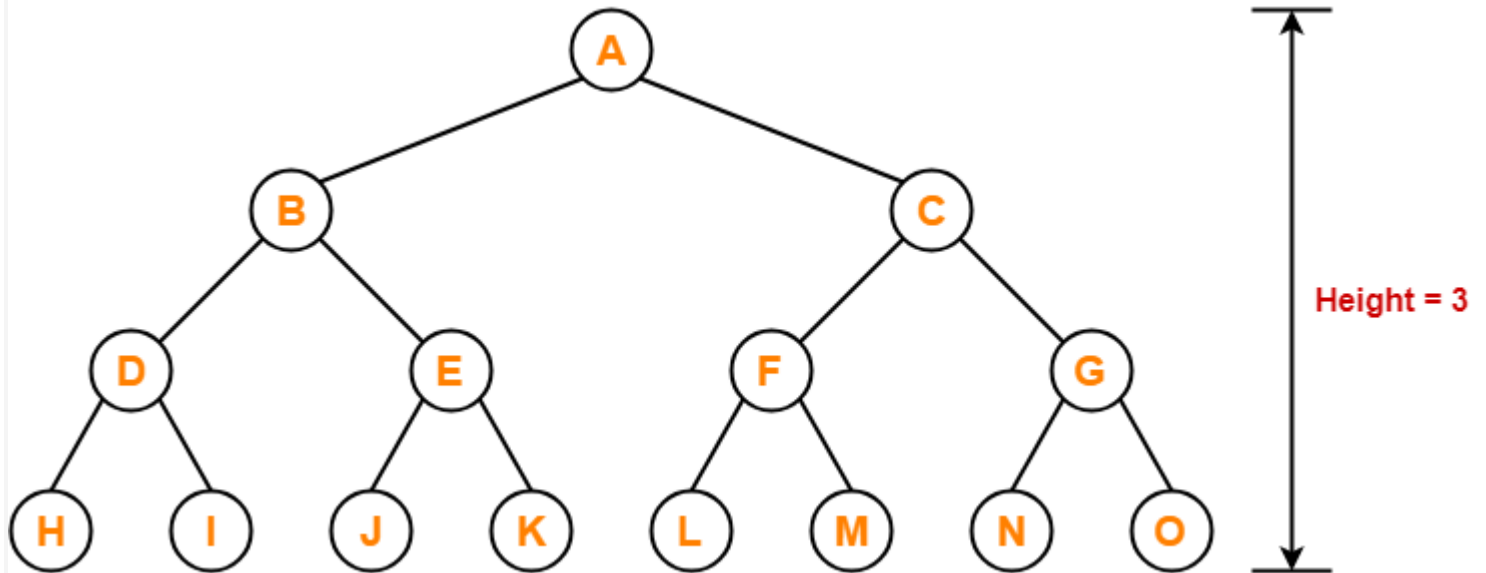
Maximum possible number of nodes in AVL tree of height-3

$$= 2^{3+1} - 1$$

$$= 16 - 1$$

$$= 15$$

Thus, in AVL tree of height-3, maximum number of nodes that can be inserted = 15.



We can not insert more number of nodes in this AVL tree.

Property-02:

n number of nodes in AVL Tree of height H is given by a recursive relation-

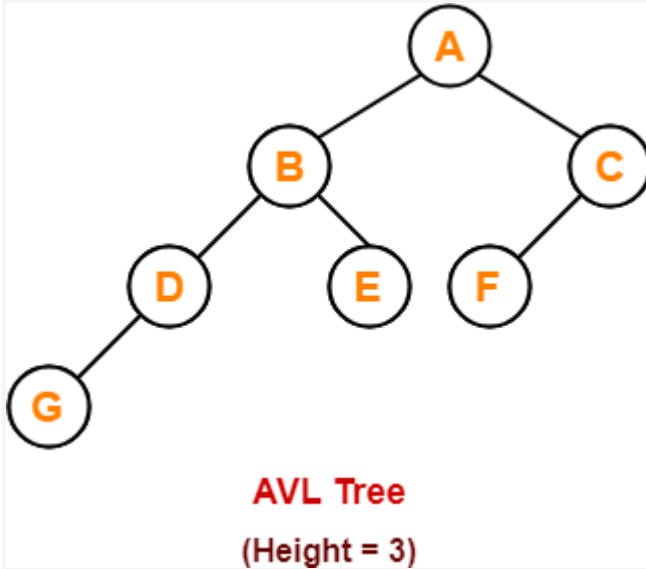
$$N(H) = N(H-1) + N(H-2) + 1$$

Base conditions for this recursive relation are-

- $N(0) = 1$
- $N(1) = 2$

Example-

Minimum possible number of nodes in AVL tree of height-3 = 7



(For explanation, Refer problem-01)

Property-03:

Minimum possible height of AVL Tree using N nodes

$$= \lfloor \log_2 N \rfloor$$

Example-

Minimum possible height of AVL Tree using 8 nodes

$$= \lfloor \log_2 8 \rfloor$$

$$= \lfloor \log_2 2^3 \rfloor$$

$$= \lfloor 3 \log_2 2 \rfloor$$

$$= \lfloor 3 \rfloor$$

= 3

Property-04:

Minimum height of AVL Tree using N nodes is calculated using recursive relation-

$$N(H) = N(H-1) + N(H-2) + 1$$

Base conditions for this recursive relation are-

- $N(0) = 1$
- $N(1) = 2$

NOTE-

- If there are n nodes in AVL Tree, its maximum height can not exceed $1.44\log_2 n$.
- In other words, Worst case height of AVL Tree with n nodes = $1.44\log_2 n$.

PRACTICE PROBLEMS BASED ON AVL TREE PROPERTIES-

Problem-01:

Find the minimum number of nodes required to construct AVL Tree of height = 3.

Solution-

We know, minimum number of nodes in AVL tree of height H is given by a recursive relation-

$$N(H) = N(H-1) + N(H-2) + 1$$

where $N(0) = 1$ and $N(1) = 2$

Step-01:

Substituting $H = 3$ in the recursive relation, we get-

$$N(3) = N(3-1) + N(3-2) + 1$$

$$N(3) = N(2) + N(1) + 1$$

$$N(3) = N(2) + 2 + 1 \quad \text{(Using base condition)}$$

$$N(3) = N(2) + 3 \quad \text{.....(1)}$$

To solve this recursive relation, we need the value of $N(2)$.

Step-02:

Substituting $H = 2$ in the recursive relation, we get-

$$N(2) = N(2-1) + N(2-2) + 1$$

$$N(2) = N(1) + N(0) + 1$$

$$N(2) = 2 + 1 + 1 \quad \text{(Using base conditions)}$$

$$\therefore N(2) = 4 \quad \text{.....(2)}$$

Step-03:

Using (2) in (1), we get-

$$N(3) = 4 + 3$$

$$\therefore N(3) = 7$$

Thus, minimum number of nodes required to construct AVL tree of height-3 = 7.

Problem-02:

Find the minimum number of nodes required to construct AVL Tree of height = 4.

Solution-

We know, minimum number of nodes in AVL tree of height H is given by a recursive relation-

$$N(H) = N(H-1) + N(H-2) + 1$$

where $N(0) = 1$ and $N(1) = 2$

Step-01:

Substituting $H = 4$ in the recursive relation, we get-

$$N(4) = N(4-1) + N(4-2) + 1$$

$$N(4) = N(3) + N(2) + 1 \quad \dots\dots\dots(1)$$

To solve this recursive relation, we need the value of $N(2)$ and $N(3)$.

Step-02:

Substituting $H = 2$ in the recursive relation, we get-

$$N(2) = N(2-1) + N(2-2) + 1$$

$$N(2) = N(1) + N(0) + 1$$

$$N(2) = 2 + 1 + 1 \quad (\text{Using base conditions})$$

$$\therefore N(2) = 4 \quad \dots\dots\dots(2)$$

Step-03:

Substituting $H = 3$ in the recursive relation, we get-

$$N(3) = N(3-1) + N(3-2) + 1$$

$$N(3) = N(2) + N(1) + 1$$

$$N(3) = 4 + 2 + 1 \quad (\text{Using (2) and base condition})$$

$$\therefore N(3) = 7 \quad \dots\dots\dots(3)$$

Step-04:

Using (2) and (3) in (1), we get-

$$N(4) = 7 + 4 + 1$$

$$\therefore N(4) = 12$$

Thus, minimum number of nodes required to construct AVL tree of height-4 = 12.

Problem-03:

What is the maximum height of any AVL tree with 10 nodes?

Solution-

For calculating the maximum height of AVL tree with n nodes, we use a recursive relation-

$$N(H) = N(H-1) + N(H-2) + 1$$

Step-01:

Substituting $H = 2$ in the recursive relation, we get-

$$N(2) = N(2-1) + N(2-2) + 1$$

$$N(2) = N(1) + N(0) + 1$$

$$N(2) = 2 + 1 + 1 \quad \text{(Using base conditions)}$$

$$\therefore N(2) = 4 \quad \text{.....(1)}$$

So, minimum number of nodes required to construct AVL tree of height-2 = 4.

Step-02:

Substituting $H = 3$ in the recursive relation, we get-

$$N(3) = N(3-1) + N(3-2) + 1$$

$$N(3) = N(2) + N(1) + 1$$

$$N(3) = 4 + 2 + 1 \quad \text{(Using (1) and base condition)}$$

$$\therefore N(3) = 7 \quad \text{.....(2)}$$

So, minimum number of nodes required to construct AVL tree of height-3 = 7.

Step-03:

Substituting $H = 4$ in the recursive relation, we get-

$$N(4) = N(4-1) + N(4-2) + 1$$

$$N(4) = N(3) + N(2) + 1$$

$$N(4) = 7 + 4 + 1 \quad (\text{Using (1) and (2)})$$

$$\therefore N(4) = 12$$

So, minimum number of nodes required to construct AVL tree of height-4 = 12.

But given number of nodes = 10 which is less than 12.

Thus, maximum height of AVL tree that can be obtained using 10 nodes = 3.

Problem-04:

What is the maximum height of any AVL tree with 77 nodes?

Solution-

For calculating the maximum height of AVL tree with n nodes, we use a recursive relation-

$$N(H) = N(H-1) + N(H-2) + 1$$

Step-01:

Substituting $H = 2$ in the recursive relation, we get-

$$N(2) = N(2-1) + N(2-2) + 1$$

$$N(2) = N(1) + N(0) + 1$$

$$N(2) = 2 + 1 + 1 \quad (\text{Using base conditions})$$

$$\therefore N(2) = 4 \quad \dots\dots\dots(1)$$

So, minimum number of nodes required to construct AVL tree of height-2 = 4.

Step-02:

Substituting $H = 3$ in the recursive relation, we get-

$$N(3) = N(3-1) + N(3-2) + 1$$

$$N(3) = N(2) + N(1) + 1$$

$$N(3) = 4 + 2 + 1 \quad (\text{Using (1) and base condition})$$

$$\therefore N(3) = 7 \quad \dots\dots\dots(2)$$

So, minimum number of nodes required to construct AVL tree of height-3 = 7.

Step-03:

Substituting $H = 4$ in the recursive relation, we get-

$$N(4) = N(4-1) + N(4-2) + 1$$

$$N(4) = N(3) + N(2) + 1$$

$$N(4) = 7 + 4 + 1 \quad (\text{Using (1) and (2)})$$

$$\therefore N(4) = 12 \quad \dots\dots\dots(3)$$

So, minimum number of nodes required to construct AVL tree of height-4 = 12.

Step-04:

Substituting $H = 5$ in the recursive relation, we get-

$$N(5) = N(5-1) + N(5-2) + 1$$

$$N(5) = N(4) + N(3) + 1$$

$$N(5) = 12 + 7 + 1 \quad (\text{Using (2) and (3)})$$

$$\therefore N(5) = 20 \quad \dots\dots\dots(4)$$

So, minimum number of nodes required to construct AVL tree of height-5 = 20.

Step-05:

Substituting $H = 6$ in the recursive relation, we get-

$$N(6) = N(6-1) + N(6-2) + 1$$

$$N(6) = N(5) + N(4) + 1$$

$$N(6) = 20 + 12 + 1 \quad (\text{Using (3) and (4)})$$

$$\therefore N(6) = 33 \quad \dots\dots\dots(5)$$

So, minimum number of nodes required to construct AVL tree of height-6 = 33.

Step-06:

Substituting $H = 7$ in the recursive relation, we get-

$$N(7) = N(7-1) + N(7-2) + 1$$

$$N(7) = N(6) + N(5) + 1$$

$$N(7) = 33 + 20 + 1 \quad (\text{Using (4) and (5)})$$

$$\therefore N(7) = 54 \quad \dots\dots\dots(6)$$

So, minimum number of nodes required to construct AVL tree of height-7 = 54.

Step-07:

Substituting $H = 8$ in the recursive relation, we get-

$$N(8) = N(8-1) + N(8-2) + 1$$

$$N(8) = N(7) + N(6) + 1$$

$$N(8) = 54 + 33 + 1 \quad \text{(Using (5) and (6))}$$

$$\therefore N(8) = 88 \quad \text{.....(6)}$$

So, minimum number of nodes required to construct AVL tree of height-8 = 88.

But given number of nodes = 77 which is less than 88.

Thus, maximum height of AVL tree that can be obtained using 77 nodes = 7.